Cocopeat Production Process

From coconut husk to export-grade baled cocopeat

Turnkey Zhengzhou Dingli line, 1 ton/hour, final moisture 15%. Six stages, 22 stations, from raw husk intake to compressed export blocks. Station numbers below (101 to 122) follow the plant layout and equipment list, so this document reads as both a plant-floor process reference and a buyer-facing specification.

Output specification at a glance

Feedstock	Raw coconut husk (whole husk)
Line throughput	1 ton / hour of finished cocopeat
Final moisture	15% (contract specification, dryer target)
Salt content (EC)	Reduced across two wash passes to export grade. Benchmark for low-EC washed cocopeat is typically ≤0.5 mS/cm; confirm exact target per buyer (not fixed in the equipment contract).
Product format	Compressed blocks, ~30 x 30 x 12 to 15 cm, ~5 kg each, ~200 bales / hour
Co-product	Cocofiber, separated at screening, processed or sold separately
Connected load	~244 kW indicative, dominated by the 110 kW shredder and 37 kW induced draft fan

Process flow

Husk intake → shred → screen (fiber / peat split) → wash + EC reduction → dewater x2 → biomass hot-air drying to 15% → cyclone + scrubber → silo → bale into export blocks

STAGE 1 Initial processing of coconut husk

Raw husk is reduced to two streams and separated. This stage sets the feed rate for the entire line.

101 Belt Conveyor

Model: DQJ50*6000mm, width 500mm **Power:** 3 kW

Bucket-type belt with high side walls to stop spillage, arranged with a horizontal bend to save floor space. It feeds raw coconut husk continuously and evenly into the shredder. The speed of this belt effectively meters the whole line to 1 ton/hour.

102 Shredding Machine

Model: KS-1Z **Power:** 110 kW

The core size-reduction unit and the single largest power draw on the line. It tears the whole husk apart into two output streams: long cocofiber and cocopeat (the fine pith and dust). High throughput with continuous, stable operation.

103 Screening Machine

Type: vibrating screen **Power:** 3 kW

Receives the shredder discharge and screens it. Oversize long fiber (cocofiber) is retained and diverted out; the fine cocopeat passes through the screen and continues to washing and drying. Cocofiber is a saleable co-product handled on a separate line.

STAGE 2 Salt washing and EC reduction

Cocopeat carries natural salts. Washing lowers electrical conductivity (EC) so the product meets export grade. Water is supplied by the plant.

104 Belt Conveyor

Model: DQB50*8000mm, width 500mm **Power:** 4 kW

Long-run transfer belt with high capacity and low failure rate. Carries screened cocopeat from the screen toward the wash line.

105 Belt Conveyor

Model: PDJ50*8000mm, width 500mm **Power:** 4 kW

Continues the transfer into the washing area. Support frame is detachable for install and maintenance.

106 Feeding Hopper (Screw Feeder)

Model: LCS-2 **Hopper:** 2 m^3 **Power:** 2.2 kW

Acts as a buffer and meters a steady feed into the wash system through a variable-frequency screw. Fitted with water-spray nozzles that begin wetting the peat and start driving salt (EC) down before the main wash belt.

107 Salt Washing Belt Conveyor

Model: DQB50*10000mm **Power:** 3 kW

Belt fitted with spray nozzles along its length. Washes the cocopeat, dissolving and flushing out sodium and potassium salts to bring EC down toward the export target and lift product quality.

STAGE 3 Dewatering

Mechanical pressing removes free water before thermal drying. Every kg of water squeezed out here is fuel the dryer does not have to burn.

108 Belt Type Dehydration Machine

Model: DLGT2000P3, belt width 2000mm **Power:** 11 kW main + 3 kW feed (14 kW)

First mechanical dewatering. Presses free water out of the washed peat, cutting the moisture load the dryer must handle and lowering drying energy cost. Heavy-duty pre-dehydration built for high-moisture material.

109 Salt Washing Belt Conveyor

Model: DQB50*10000mm **Power:** 3 kW

A second wash pass to confirm EC meets the export specification before the material is committed to drying.

110 Belt Type Dehydration Machine

Model: DLGT2000P3 **Power:** 11 kW main + 3 kW feed (14 kW)

Second dewatering pass. Brings mechanical moisture down as far as pressing can, so the thermal dryer only has to remove the remaining water.

STAGE 4 Drying

The heart of the line. Biomass-fired hot air dries the peat down to the contract target of 15% moisture.

111 Screw Conveyor

Model: LXSU300, L=6000mm **Power:** 5.5 kW

Meters the dewatered peat from the presses into the dryer inlet at a controlled rate.

112 Biomass Hot Air Furnace

Model: RFL-240 **Output:** ~2.4 million kcal/hour **Aux power:** ~5.5 kW

Biomass-fired furnace that produces the clean hot air for drying. This is the line's main energy source. Running on biomass (coconut residue and similar) keeps drying cost low versus diesel or gas. Includes the furnace body and a dust settling chamber; wide, stable hot-air temperature range.

113 Hot Air Duct

Model: HD-12, Φ1200 x 2000mm **Power:** passive (no motor)

Insulated duct that carries hot air from the furnace to the dryer inlet. Bolted connections for straightforward install.

114 Three Layer Rotary Dryer

Model: DLTG-2110/3 **Drum:** Ø2.1m x 10m, three drums **Power:** 11 kW drive

The main drying unit. Wet peat and hot air enter the front end together; as the drum turns, material tumbles through three concentric passes, maximizing heat and mass transfer. Moisture evaporates and leaves with the exhaust to the cyclone. Product exits at the target 15% moisture. Insulated drum with 0.5mm color-steel cladding.

STAGE 5 Dust and emission control

Captures product fines and cleans the exhaust so the plant meets environmental standards. This train also keeps airflow through the dryer stable.

115 Cyclone Dust Collector + Air Locker

Cyclone: DXF180 (passive) **Air locker:** YCD-26, 2.2 kW

The cyclone spins dried peat particles and fines out of the exhaust airstream by centrifugal force, recovering product with low maintenance. The air locker (rotary valve) at the bottom discharges the collected material without breaking the system's air pressure or seal.

116 Induced Draft Fan

Model: Y9-38-9D **Power:** 37 kW **Pressure:** 2900 to 3000 Pa, VFD

The main suction fan and the second-largest power draw. It pulls hot air through the dryer, holds airflow steady, and directly drives drying efficiency. Speed is controlled by variable-frequency drive.

117 Wet Dust Collector (Scrubber)

Model: SC20 **Power:** 7.5 kW

Final emission stage. Captures the remaining fine dust from the exhaust using a water scrubber and a recirculating water pit, cutting air pollution and helping meet environmental limits. The recirculating pit is built on site to the supplier's drawings.

STAGE 6 Packing and baling

Dry cocopeat is buffered and pressed into standard export blocks. The block format is what makes cocopeat cheap to ship.

118 Screw Conveyor

Model: LXSU300, L=5000mm **Power:** 5.5 kW

Moves dried cocopeat from the dryer and cyclone discharge toward the packing area.

119 Skirted Belt Conveyor

Model: PDJ50*8000mm, width 500mm **Power:** 4 kW

Transfers and elevates the dried peat into the packing silo. Skirted edges contain the light, dry material so it does not blow off the belt.

120 Packing Silo

Model: LC-5 **Power:** passive (gravity feed)

Buffer hopper that holds finished dry cocopeat above the baler and feeds it by gravity, decoupling the dryer from the press cycle.

121 Cocopeat Baler (Block Press)

Model: YWK2-250 **Force:** 1000 kN **Power:** 5.5 kW

Compresses dry cocopeat into export blocks. Block size is roughly 30 x 30 x 12 to 15 cm, about 5 kg each, at around 200 bales/hour, which lines up with the 1 ton/hour line output (200 x 5 kg = 1000 kg). Blocks are the standard export format: compact, stackable, and low freight cost per kg.

STAGE – Control system

122 Electrical Control Cabinet

Type: XL standard cabinet **Includes:** VFD / inverters, temperature control

Central control for the whole plant from a single operating point. It runs and interlocks every motor and conveyor, houses the variable-frequency drives for speed control, and manages dryer temperature. This is where the line is started, stopped, and tuned.

Equipment and power summary

Full station list with model and connected power. “Passive” means no drive motor (ducting, cyclone body, silo). Connected load is indicative at roughly 244 kW; actual running load is lower because not every motor peaks at once.

NO.	EQUIPMENT	MODEL / SPEC	KW	STAGE
101	Belt conveyor (bucket type)	DQJ50*6000, W500	3	1 · Prep
102	Shredding machine	KS-1Z	110	1 · Prep
103	Screening machine	vibrating screen	3	1 · Prep
104	Belt conveyor	DQB50*8000, W500	4	2 · Wash
105	Belt conveyor	PDJ50*8000, W500	4	2 · Wash
106	Feeding hopper (screw feeder)	LCS-2, 2 m³	2.2	2 · Wash
107	Salt washing belt conveyor	DQB50*10000	3	2 · Wash
108	Belt dehydration machine	DLGT2000P3	14	3 · Dewater
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111	Screw conveyor	LXSU300, L=6000	5.5	4 · Dry
112	Biomass hot air furnace	RFL-240	~5.5	4 · Dry
113	Hot air duct	HD-12, Φ1200	passive	4 · Dry
114	Three layer rotary dryer	DLTG-2110/3	11	4 · Dry
115a	Cyclone dust collector	DXF180	passive	5 · Emission
115b	Air locker (rotary valve)	YCD-26	2.2	5 · Emission
116	Induced draft fan (VFD)	Y9-38-9D	37	5 · Emission
117	Wet dust collector (scrubber)	SC20	7.5	5 · Emission
118	Screw conveyor	LXSU300, L=5000	5.5	6 · Pack
119	Skirted belt conveyor	PDJ50*8000, W500	4	6 · Pack
120	Packing silo	LC-5	passive	6 · Pack
121	Cocopeat baler (block press)	YWK2-250, 1000 kN	5.5	6 · Pack
122	Electrical control cabinet	XL standard, VFD	control	Line

Utilities and site requirements

Power. ~244 kW connected load (indicative). Dominated by the 110 kW shredder and 37 kW induced draft fan.

Fuel. Biomass for the hot air furnace (coconut residue and similar), keeping drying cost low.

Water. Plant-supplied. Needed for the wash belts, the humidifying hopper, and the wet scrubber's recirculating pit.

Civil works. Equipment foundations and ground hardening are the plant's responsibility. The scrubber's recirculating water pit is built on site to the supplier's drawings.

Commissioning. One supplier engineer guides installation, commissioning, and operator training over 40 days on site. Installation labor is provided by the plant. Factory acceptance test (FAT) runs before shipment; site acceptance test (SAT) runs after install to confirm the line hits the agreed parameters.

Prepared by AIS. Station numbering and specifications follow the equipment layout and supply contract for the production line. Moisture (15%) is contractual; EC target is buyer-specified and should be confirmed against each buyer's acceptance spec.